TECHYUVA 11.0 • ANNUAL TECHNICAL SYMPOSIUM

BUDDHA DEGREE COLLEGE, GIDA, GORAKHPUR

Event Date: 3rd to 5th November, 2026

Dynamic Cointegration-Based Statistical Arbitrage in Cryptocurrency Markets

High-Frequency Algorithmic Pairs Trading with Zero-Lookahead Simulation & Live Paper OMS

Presenter: Vikash & Research Team

Course / Branch: B.Tech / B.Sc (Computer Science / Data Science)

Semester: Final Year | Department of Computer Science & Applications

Algorithmic Trading & Quantitative Finance

Table of Contents

01. Introduction

Core concepts of Statistical Arbitrage and market-neutral pairs trading in crypto

02. Case Situation

24/7 continuous trading universe & 8,640 5-minute candles high-frequency dataset

03. Key Challenge

Non-stationarity, spurious correlations, multiple testing bias, and transaction fees

04. Company / Quant Strategy

Engle-Granger testing, FDR Benjamini-Hochberg correction, and Kalman filters

05. Results & Impact

30-day performance benchmark: +2.90% Net Return, 20.62 Sharpe, 96.1% Win Rate

06. Analysis

Why Static OLS outperformed Dynamic Kalman Filter under microstructure friction

07. Key Learnings

Mathematical discipline, execution latency management, and full-stack deployment

08. Conclusion & Demo

Production-grade React terminal, zero-secret paper OMS, and GitHub architecture

Introduction: Algorithmic Statistical Arbitrage

What is Statistical Arbitrage?

- **Market-Neutral Pairs Trading**
  Simultaneously buying an undervalued asset while shorting an overvalued asset to hedge out broader market risk.
- **Cointegration vs. Correlation**
  Correlation measures co-movement in returns. Cointegration establishes a true stationary long-term equilibrium: $S_t = \ln(P_A) - \alpha - \beta \ln(P_B)$.
- **Mean-Reversion Law**
  When the spread diverges beyond ±2.0 standard deviations (Z-score), statistical forces pull it back, yielding predictable alpha.
- **Zero Directional Risk**
  Profits are generated whether Bitcoin is rallying or crashing.

System Scope & Deliverables

- **Automated Screening Engine**
  Evaluates all 45 cross-pairs in Binance USDT-M Futures using Engle-Granger two-step tests.
- **Multiple Hypothesis Correction**
  Applies Benjamini-Hochberg False Discovery Rate (FDR) control at $q \leq 0.05$ to eliminate false discoveries.
- **Zero-Lookahead Backtester**
  Simulates realistic order execution at bar t+1 Open with exchange taker fees (0.05%) and slippage (0.02%).
- **Production Web Terminal**
  React 18 frontend + FastAPI backend + local paper trading engine without needing private API secret keys.

Case Situation: Market Environment & Dataset

Crypto Microstructure

- Continuous 24/7/365 trading with no opening/closing bell or market halts.
- Extreme volatility and frequent leverage liquidation cascades.
- Orderbook depth shifts rapidly during macro events.
- Traditional equity stat-arb assumptions fail in crypto due to fat-tailed returns.

Empirical Dataset

- Venue: Binance USDT-M Perpetual Futures (deepest global liquidity pool).
- Asset Universe (10): BTC, ETH, SOL, BNB, XRP, ADA, LINK, DOGE, AVAX, LTC.
- Frequency: 5-minute continuous OHLCV candles.
- Horizon: 8,640 consecutive synchronized bars (30 full trading days).
- Cross-Pairs: 45 unique combinations screened.

Execution Realism

- Initial Portfolio: $10,000 USDT capital base.
- Pair Sizing: 10% allocation ($1,000 target gross exposure per pair).
- Exchange Taker Fee: 0.05% per leg (0.10% round-trip friction).
- Execution Slippage: 0.02% modeled adverse fill on both entry and exit.
- Signal Integrity: Computed at Bar t Close, filled strictly at Bar t+1 Open.

1. Spurious Correlation & Non-Stationarity

Raw crypto prices are non-stationary I(1) random walks. High correlation does NOT imply mean-reversion. Trading non-cointegrated pairs leads to permanent capital divergence.

2. Data Snooping in Multi-Pair Screening

Testing 45 candidate pairs at $\alpha = 0.05$ yields an expected ~2.25 false discoveries purely by chance. Overfitting to past noise leads to catastrophic breakdown out-of-sample.

3. Microstructure Friction & Fee Drag

In 5-minute high-frequency trading, small paper profits are easily eaten by 0.05% taker fees and 0.02% slippage unless trade frequency and profit targets are strictly calibrated.

4. Kalman Filter Over-Adaptation in Micro-Noise

Adaptive state-space estimation can mistake random tick noise for structural shifts in hedge ratio β, triggering excessive turnover and severe whipsaw losses.

Step 1: Ingestion & Alignment

Download continuous 5m OHLCV tick data from Binance, verify zero missing candles, and compute log prices $X_t = \ln(P_A)$, $Y_t = \ln(P_B)$.

Step 2: Engle-Granger Cointegration

Run OLS regression $X_t = \alpha + \beta Y_t + \varepsilon_t$. Test residual stationarity via Augmented Dickey-Fuller (ADF) with constrained lag selection.

Step 3: Benjamini-Hochberg FDR Control

Sort p-values across all 45 pairs: $q_{(k)} = \min(m/k * p_{(k)}) \leq 0.05$. Rejects 80%+ spurious pairs to ensure statistically proven cointegration.

Step 4: Half-Life Mean Reversion Filter

Fit Ornstein-Uhlenbeck process $\Delta S_t = \lambda S_{(t-1)} + \eta_t$. Keep only pairs with fast reversion: $HL = -\ln(2)/\lambda \leq 24.0$ hours.

Step 5: Dynamic State Machine & Sizing

Compute rolling Z-score: $Z_t = (S_t - \mu) / \sigma$. Enter Long Spread at $Z \leq -2.0$, Short Spread at $Z \geq +2.0$, exit at Z crossing 0.0. Fixed $1k sizing.

Step 6: Live Paper OMS & Web UI

Real-time order fill simulation with taker fees & slippage. Stream live status to responsive React 18 & FastAPI terminal.

Results & Impact: Empirical Benchmark Performance

NET RETURN (30D)
+2.90%
Annualized: +41.6%

SHARPE RATIO
20.62
Risk-Adjusted Alpha

MAX DRAWDOWN
-0.11%
Near-Zero Downside

WIN RATE
96.05%
73 Wins / 3 Losses

Model Benchmark Comparison: Static OLS vs. Dynamic Kalman Filter

Performance Metric	Model M1: Static OLS	Model M3: Dynamic Kalman Filter	Strategy Edge & Significance
Total Net Return / Profit	+$290.22 (+2.90%)	-$389.38 (-3.89%)	+6.79% alpha spread in favor of Static OLS
Annualized Sharpe Ratio	20.62	-45.91	Exceptional risk-adjusted return under static cointegration
Maximum Drawdown	-0.11%	-3.89%	Zero severe drawdown regimes observed in Model M1
Win Rate (% Trades)	96.05% (73 of 76)	18.45% (62 of 336)	M1 selectively trades only high-probability extremes
Profit Factor	30.66	0.10	Asymmetric profit distribution in favor of static model
Average Holding Duration	2.15 hours (25.8 bars)	0.58 hours (7.0 bars)	Static model aligns with true Ornstein-Uhlenbeck half-life
Total Friction Paid	$108.13	$461.84	Kalman over-traded 4.4x more, destroying equity via fee drag

Quantitative Analysis: The Kalman Overfitting Paradox

Why Static OLS Succeeded (M1)

✔ Stable Mathematical Anchor

Calibrated parameters ($\alpha = 1.4648$, $\beta = 1.1678$) reflected the structural cointegrating relationship between BTC and ETH across the full 30 days.

✔ Patience in Entry Execution

Waited strictly for large $\pm 2.0\sigma$ deviations, entering only 76 high-probability round-trips over 8,640 bars.

✔ Natural Mean-Reversion

Gave trades sufficient time to revert (average holding duration = 2.15 hours), matching the empirical half-life of 1.19 hours.

✔ Minimal Fee Overhead

Kept total transaction costs to just $108.13, preserving over 73% of gross trading profits.

Why Dynamic Kalman Failed (M3)

✘ Over-Adapting to Micro-Noise

At 5-minute frequency, the Kalman filter continuously shifted β in response to temporary orderbook imbalances rather than true economic regime shifts.

✘ The Moving Goalpost Trap

As the spread expanded, the Kalman filter adjusted β to absorb the divergence, prematurely moving the equilibrium and closing trades at false inflection points.

✘ Severe Turnover Explosion

Executed 336 round-trip trades (4.4x more than static), creating continuous churning and whipsaws.

✘ Catastrophic Fee Drag

Accumulated $461.84 in taker fees and slippage, wiping out capital and yielding a -45.91 Sharpe ratio.

Key Learnings & Quant Insights

1. Complexity Does Not Guarantee Alpha

Sophisticated time-varying algorithms (like Kalman Filters) often underperform simpler static regressions in noisy high-frequency regimes due to parameter variance and fee multiplication.

2. Execution Friction Dominates Strategy PnL

Theoretical paper returns without modeled taker fees (0.05%) and slippage (0.02%) are completely misleading. Real-world statistical arbitrage is an optimization against transaction costs.

3. Statistical Hygiene Prevents Overfitting

Controlling the False Discovery Rate (FDR) via Benjamini-Hochberg is indispensable when mining hundreds of asset combinations to avoid trading random statistical artifacts.

4. Full-Stack Systems Engineering

Building a real quant system requires uniting mathematical modeling (NumPy/Pandas/Statsmodels), backtesting integrity, async API backends (FastAPI), and responsive monitoring (React 18).

Conclusion: Summary of Achievements

✔ **Research-Grade Quantitative Rigor**

Engineered a robust pairs trading framework grounded in Engle-Granger cointegration, Augmented Dickey-Fuller tests, and Benjamini-Hochberg FDR control.

✔ **Proven Empirical Edge**

Achieved +2.90% net 30-day return (41.6% annualized) with an exceptional 20.62 Sharpe ratio and 96.05% win rate on BTC/ETH futures, including realistic trading costs.

✔ **Solved Zero-Lookahead Bias**

All signals evaluated strictly at candle close and filled at next-bar open, preventing lookahead leakage present in naive backtesters.

✔ **Production-Ready Full-Stack Software**

Created a standalone simulated Order Management System (OMS) that paper-trades live public orderbooks with zero API secret keys required.

✔ **Open Source & Cloud Deployed**

Full code repository, test suites, React terminal, and documentation publicly accessible on GitHub and GitHub Pages.

THANK YOU!

Questions & Answers (Q/A)

TECHYUVA 11.0 | BUDDHA DEGREE COLLEGE, GIDA, GORAKHPUR

GitHub: https://github.com/vikash21pa-wq/Dynamic_Cointegration-Based_Statistical-_arbitrage

Live Web Terminal: https://vikash21pa-wq.github.io/Dynamic_Cointegration-Based_Statistical-_arbitrage/

Presented by: Vikash & Research Team (Final Year CS/DS)